

AI Inference Engineering and Systems Design · Final project

# A learning companion on a scarce GPU

An agentic RAG app on our own vLLM cluster on Lambda GPUs. Admission control decides what enters, and llm-d decides where it goes. vLLM runs the model.

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2026-10-10

[github.com/chetananand/learning-companion-serving](https://github.com/chetananand/learning-companion-serving)

# The app: a learning companion over my bookmarks

My Notion database has 998 bookmarks of pages that I want to learn from. The companion answers my questions from these bookmarks, and it cites a source for each fact.

Quick mode: one agent searches the bookmarks, reads a page, and reads a figure with OCR.

Verified mode: bookmarks get old, so a second agent checks up to 3 claims on live web pages. Our code decides when a live fact wins over a bookmark.

## In course terms

Track B with a Track A tool: an agent that retrieves.

▶ **Clip 1, 27 s: a verified answer, both claims verified**

## I ask a question

in the chat of the app



## One agent finds the facts

search the bookmarks, read a page, read a figure



## The answer

with a source for each fact

## ↓ verified mode only

## A second agent checks the claims

up to 3 claims, on live web pages



## The final answer

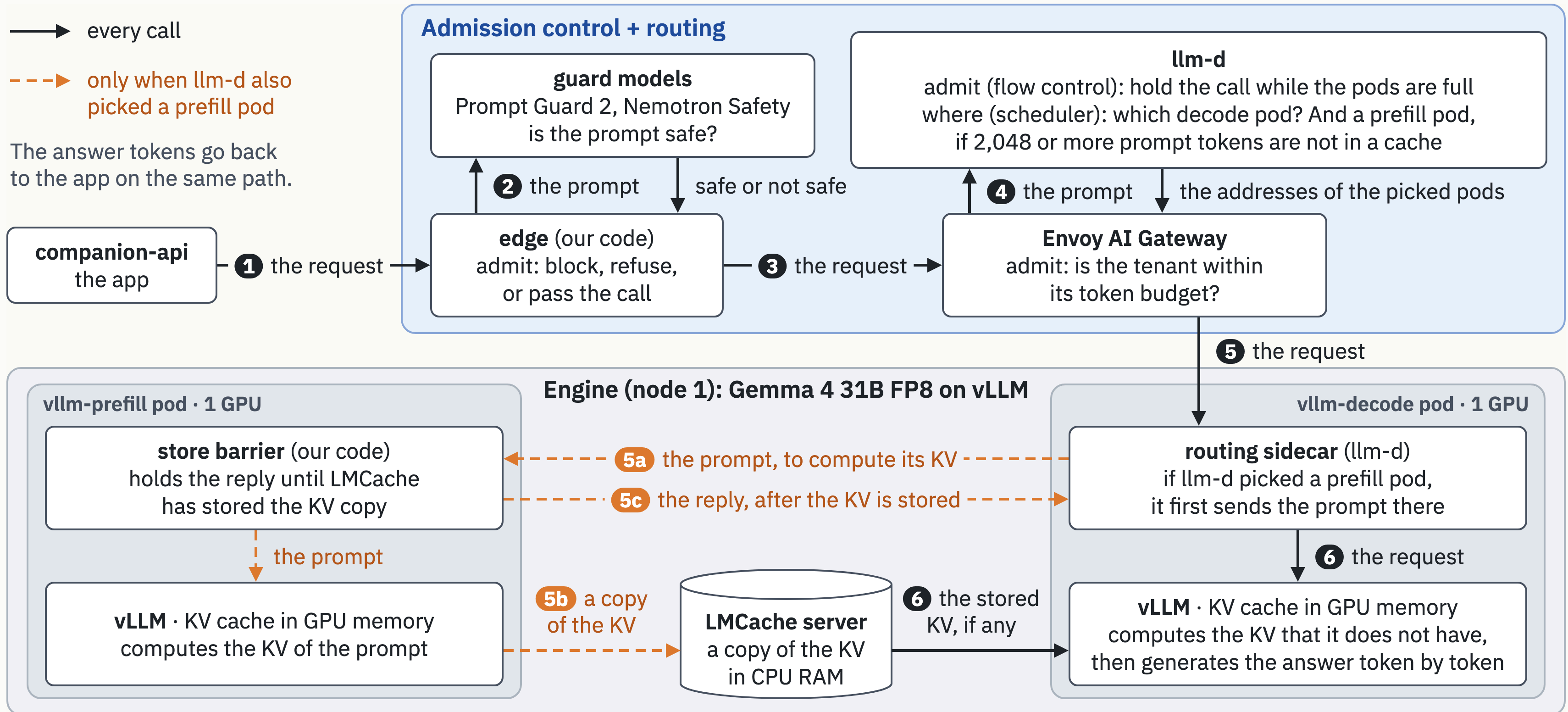
with the status of each claim

# The path of one LLM call

→ every call

---> only when llm-d also picked a prefill pod

The answer tokens go back to the app on the same path.



# The models, and the job of each

| Job                                | Model                                           | Where it runs                                             |
|------------------------------------|-------------------------------------------------|-----------------------------------------------------------|
| The answers and all agent steps    | Gemma 4 31B, FP8 (RedHatAI), on vLLM v0.30.0    | node 1: the prefill pod and the decode pod, one H100 each |
| Prompt injection check             | Llama Prompt Guard 2, 86M                       | node 2: a GPU slice of 4,000 MiB                          |
| Content safety check               | Nemotron 3.5 Content Safety, in NeMo Guardrails | node 2: a GPU slice of 16,000 MiB                         |
| Embeddings for the bookmark search | BAAI bge-m3                                     | node 2: SIE, a GPU slice of 12,000 MiB                    |
| Rerank of the search results       | Qwen3 Reranker 4B                               | node 2: SIE, the same slice                               |
| Text in figures (OCR)              | LightOnOCR-2, 1B                                | node 2: SIE, a GPU slice of 20,000 MiB                    |

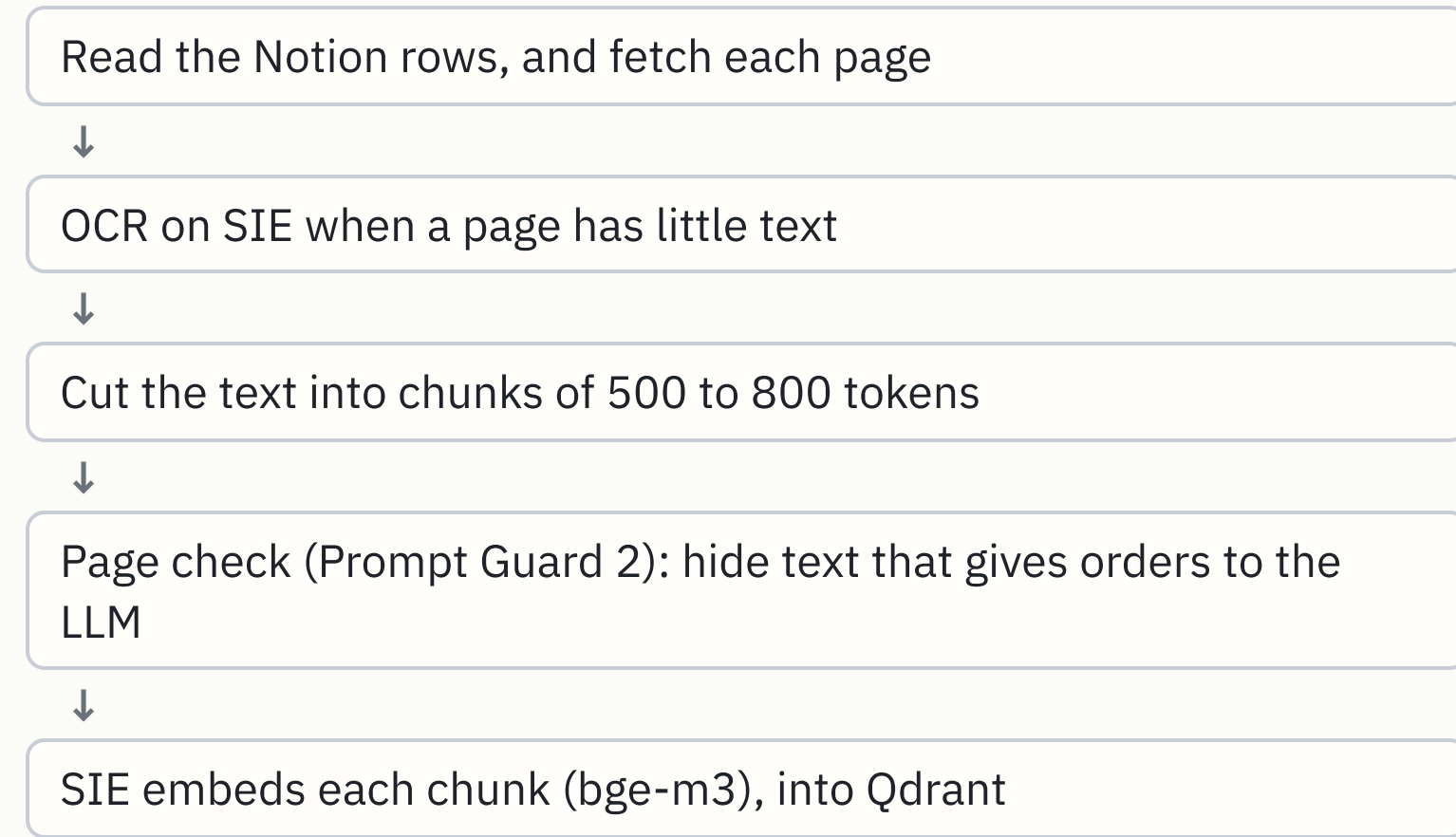
HAMi cuts one H100 of node 2 into these slices. On the A100 node, the small models shared one GPU.

## Why Gemma 4 31B

A tested FP8 checkpoint that fits one H100. It passed all 7 of our gate tests. They test tool calls (39 of 40, so 97.5%), TTFT, decode speed, the prefix cache, a P/D split, and the CPU tier. We dropped Qwen3.8-27B for its open vLLM bugs in the prefix cache.

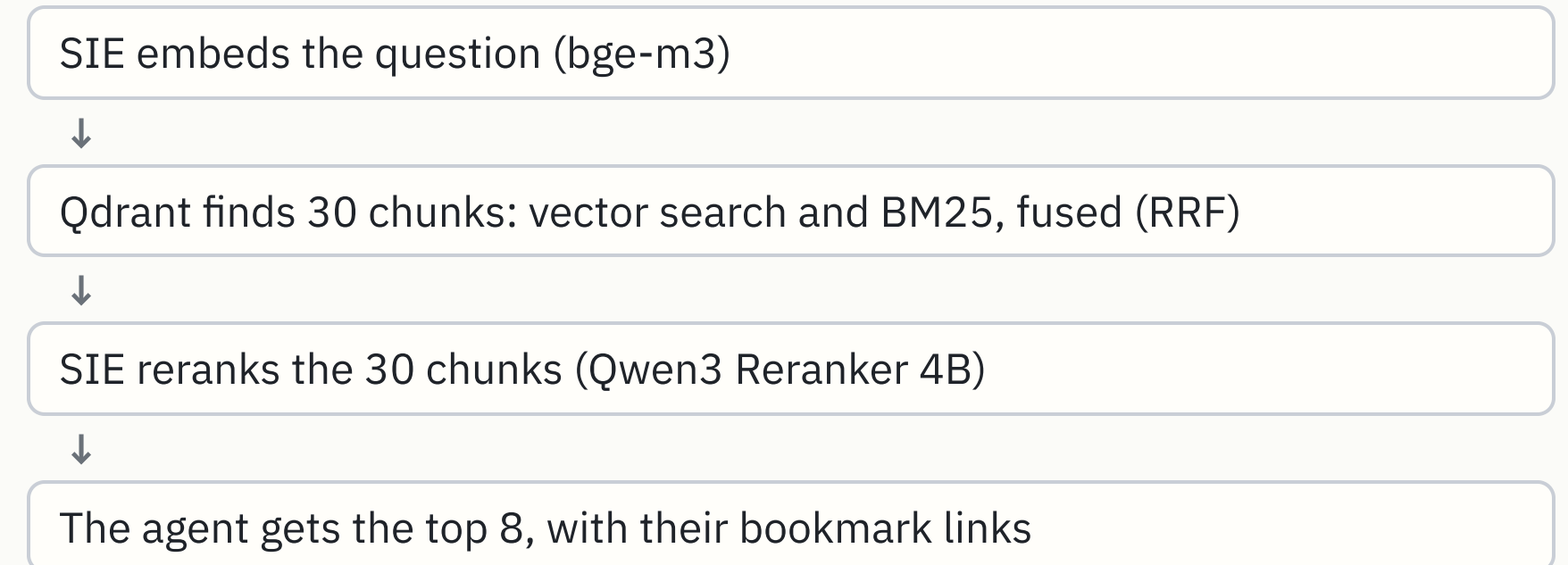
# The bookmark search: ingest once, then search in each turn

## Ingest: a job in the cluster, before all tests



It ran on 2026-09-28: 910 of 1,000 rows gave 4,112 chunks. Each later session restored a snapshot of Qdrant from the laptop.

## Search: in each turn, a tool of the agent



### SIE and Qdrant

SIE is the model server for the small models: embed, rerank, and OCR. Qdrant stores the vectors and searches them. The search makes no LLM call.



# KV on paper: bytes for each token, and how many sequences fit

Gemma 4 31B, BF16 KV. Full-attention layers: 40,960 bytes for each token. Sliding-window layers: 819,200 bytes for each token, but only for the last 1,024 tokens, so 800 MiB for each sequence.

$$\text{max seqs} = (\text{HBM} - \text{weights} - \text{activations}) / \text{KV of one sequence}$$

| Length                                    | KV of one sequence | Max seqs, BF16 KV | Max seqs, FP8 KV |
|-------------------------------------------|--------------------|-------------------|------------------|
| 5,121 tokens: what the app sends (median) | 0.98 GiB           | 36                | 72               |
| 8K tokens: the length that we planned     | 1.09 GiB           | 32                | 64               |
| 32K tokens: max_len                       | 2.03 GiB           | 17                | 34               |

One H100, the KV budget after the weights: 35.3 GiB.

## WHAT WE MEASURED

The KV cache that vLLM gave the decode pod (one H100).

173,657 tokens

with BF16 KV, and 345,235 tokens with FP8 KV

### Model switch

We kept Gemma 4 31B, because it passed all 7 gate tests. With Muse Glimmer 30B, the KV of an 8K sequence is 83% smaller, and 189 sequences fit at 8K, not 32. We changed the KV to FP8 in a test instead: it halves the bytes for each token. The base runs used BF16 KV.

Later in the talk: which limiter came first, and if our guess was right.

# The deployment: two nodes, and an A100 fallback

- Node 1 · 2 × H100 SXM 80 GB · the engine**

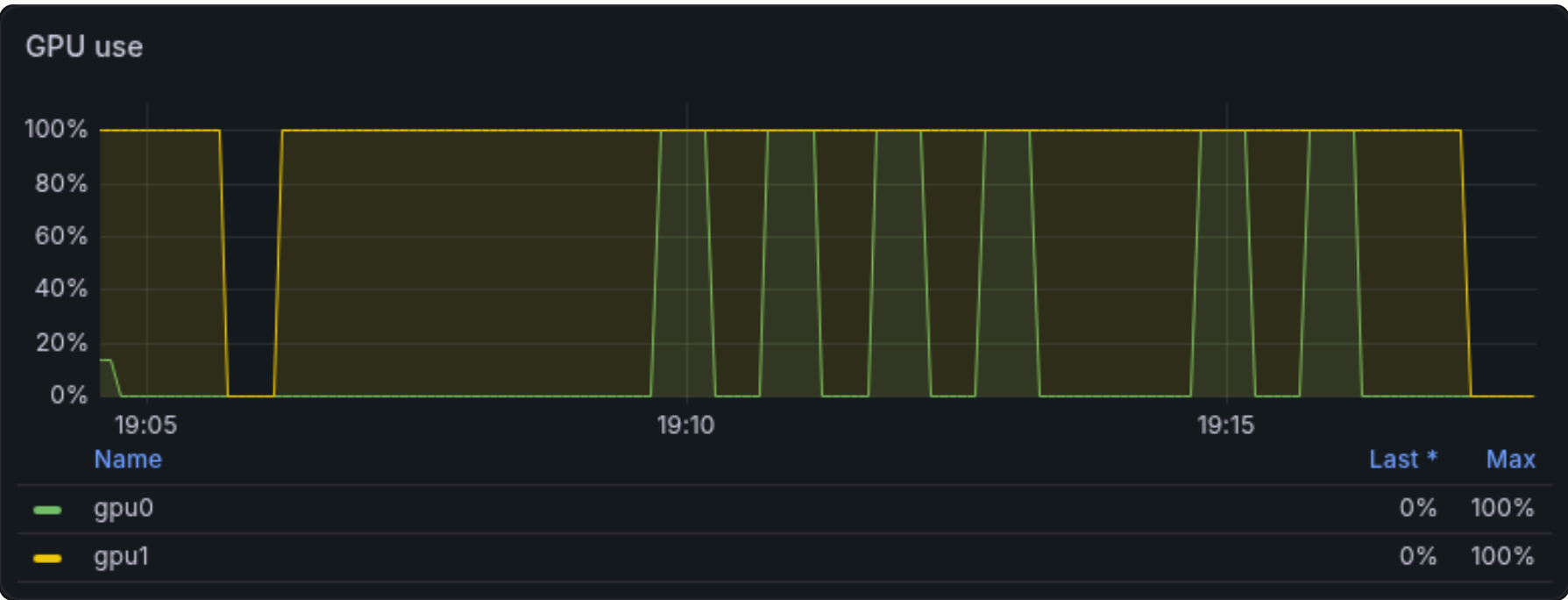
vllm-prefill on GPU 0 and vllm-decode on GPU 1, each on a full GPU. The LMCache server: up to 250 GiB of the 450 GiB of CPU RAM.
- Node 2 · 1 × H100 80 GB · control and data**

HAMi slices of one GPU for the SIE and guard models. edge, guard, Envoy, llm-d, the app, Qdrant, Redis, Prometheus, Grafana, and KEDA.
- 2026-10-01 · one node · 8 × A100 80 GB**

No H100 had stock. HAMi on all GPUs: each engine pod gets a full GPU, and the node 2 pods share one GPU.

The laptop reaches the cluster only through an SSH tunnel to ports on 127.0.0.1.

Dashboard 1 · Cluster: the use of each engine GPU at 150% load. Both GPUs are at 100%.





# The cluster design: each choice, its reason, and the proof

| Item                   | Our choice                                                                           | Why                                                                                               | Proof                                                                  |
|------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| GPU                    | H100 SXM 80 GB. Each vLLM pod gets a full GPU.                                       | The smallest GPU with FP8 compute that meets the TTFT goal. An A6000 leaves 6.8 GiB for KV.       | TTFT on paper at 8K: 0.62 s, against 3.95 s (A100) and 7.96 s (A6000). |
| Model                  | Gemma 4 31B FP8. Weights: 30.4 GiB. KV at 5,121 tokens: 0.98 GiB.                    | It passed all 7 gate tests, with 97.5% correct tool calls.                                        | The KV math (slide 6)                                                  |
| Topology               | 1 prefill pod and 1 decode pod. A call splits only at 2,048 or more uncached tokens. | Short or cached calls stay on the decode pod, with no hop. Each pool has its own scale signal.    | For our traffic, two colocated replicas were faster (slide 17).        |
| Slices                 | None for vLLM. HAMi slices one GPU for SIE and the guard models.                     | The KV needs all the free HBM. The small models need 52 GB in total.                              | The deployment (slide 7)                                               |
| Concurrency            | Decode: 24 sequences. Prefill: 8. Max length: 32,768 tokens.                         | At 8K, 32 sequences fit, and at 24K, about 20. At 24, the time between tokens stays near 20 ms.   | With 32 sequences, the TTFT p50 fell from 4.59 s to 2.10 s.            |
| Hop backend            | The LMCache server (our name: lmcache), with our store barrier                       | Gemma 4 runs on it. Mooncake: no Gemma 4 test, no RDMA on our nodes. NIXL over TCP: 4.1 to 4.3 s. | Hop TTFT: 0.52 to 0.78 s (slide 13)                                    |
| Overflow               | qwen3.8-27b on the Superlinked API, only for an interactive 503 or 529               | Quality near ours, the same API, and a Redis limiter on the cost. It was off in all runs.         | The gate marked the 124 interactive timeout calls.                     |
| Two boxes, two workers | Gateway: edge, the Envoy AI Gateway, and llm-d. Engine: two vLLM pods.               | The handout puts admit, place, and the queue in the gateway, and vLLM in the engine.              | The architecture (slide 3)                                             |
| Scale                  | KEDA, for each pool. Prefill: uncached prefill tokens. Decode: running sequences.    | Each pool grows on its own signal, so only the hot pool grows.                                    | A new decode pod 15 s after the request (slide 15)                     |



# What the app sends: short agent steps with a cached prefix

64%

of the prompt of an agent step was in the prefix cache

60%

of the calls had fewer than 2,048 new tokens

## HOW WE LOAD THE CLUSTER

We replay recorded app turns at random times. A turn is one question with all its LLM calls. 100% load is 54 turns each minute, on average. At this rate, a soak test that adds load each minute refused its first call.

## WHAT WE MEASURED

The tokens of 1,606 real app calls, from the Envoy logs of four capture runs.

| Step   | Calls | Prompt tokens (median) | Share in the prefix cache | New tokens (median) |
|--------|-------|------------------------|---------------------------|---------------------|
| quick  | 978   | 5,121                  | 0.19                      | 4,515               |
| verify | 321   | 1,427                  | 0.65                      | 133                 |
| agent  | 303   | 2,121                  | 0.64                      | 459                 |

The quick steps come from the quick agent. The agent steps come from the second agent, which picks the claims and writes the final answer. The verify steps come from its helper, which checks one claim. New tokens are the prompt tokens that were not in the prefix cache, so the engine computed them.

### What this means

The quick steps carry the bookmark chunks that the search found, so most of their tokens are new: the Track A shape. The agent steps share a long prefix: the Track B shape.

# Guard and stay or leave happen before any GPU work

```
control/edge/guard.py:105 inspect()
control/edge/overflow.py:19 should_leave()
```

## WHAT WE MEASURED

200 chat turns through the full path: 22 prompt-injection attacks and 178 normal turns.

## 22 of 22 attacks blocked

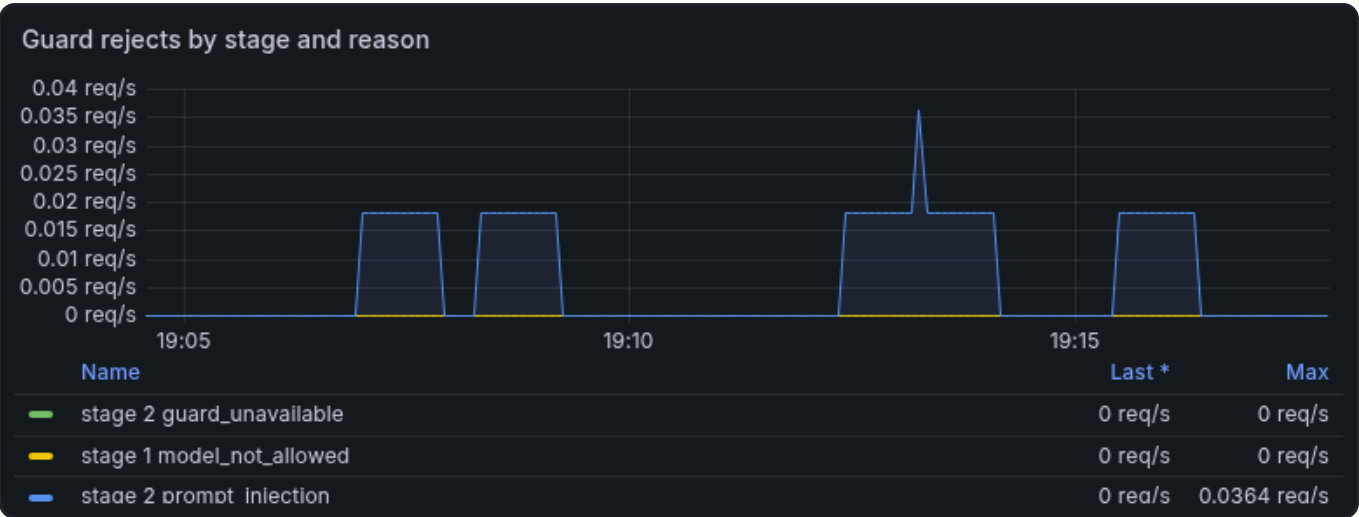
and 0 of 178 normal turns blocked. One check took 0.19 s (p50).

## WHAT WE MEASURED: STAY OR LEAVE

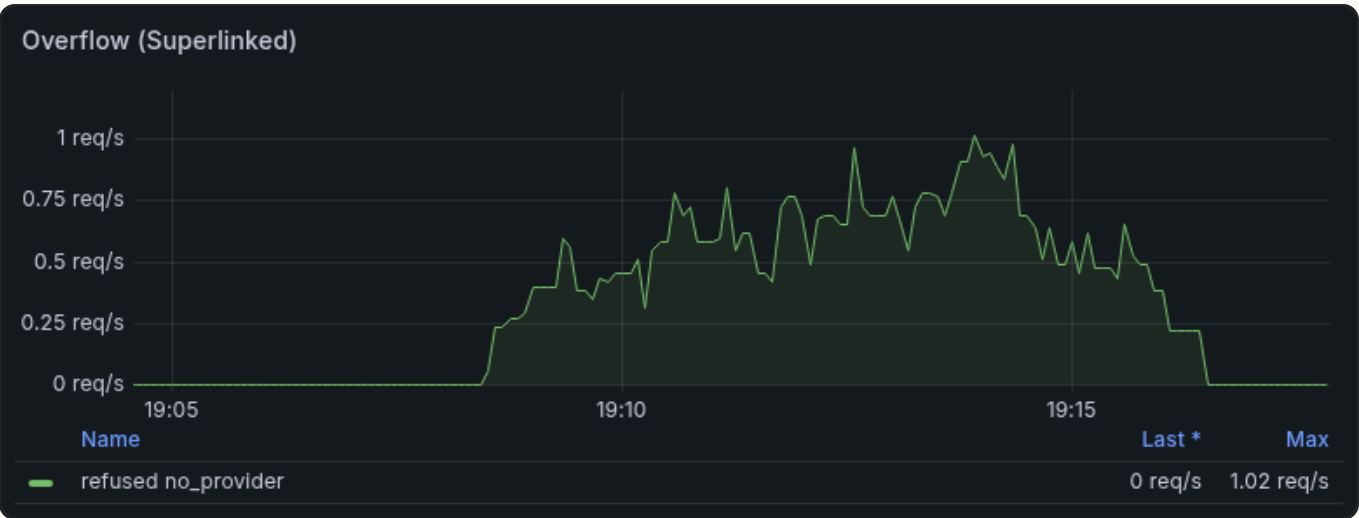
At 150% load, which calls the leave gate let go. A 429, a 500, and a batch call must stay.

The gate let only the 124 interactive 503 timeout\_queue calls go. No 429 left.

Dashboard 3 · Gateway + admission: guard rejects by stage and reason



Dashboard 2 · Success and failures: the calls that may leave. The overflow was off, so they got a 503.



guard

stay or leave

admit

place

hop

warm

scale

# Admit: we refuse work at the door, not in the engine

```
control/router/policy.yaml tenants,  
flow_control  
control/edge/admit.py:23 slice_oom
```

Each tenant has a token budget for each minute. While the pods are full (5 queued requests or 90% KV use), the llm-d flow control holds the calls.

## WHAT WE MEASURED

In all load tests, from 50% to 150% load: the preemptions in the vLLM engine.

## 0 preemptions

Before the KV was full, flow control refused the extra work.

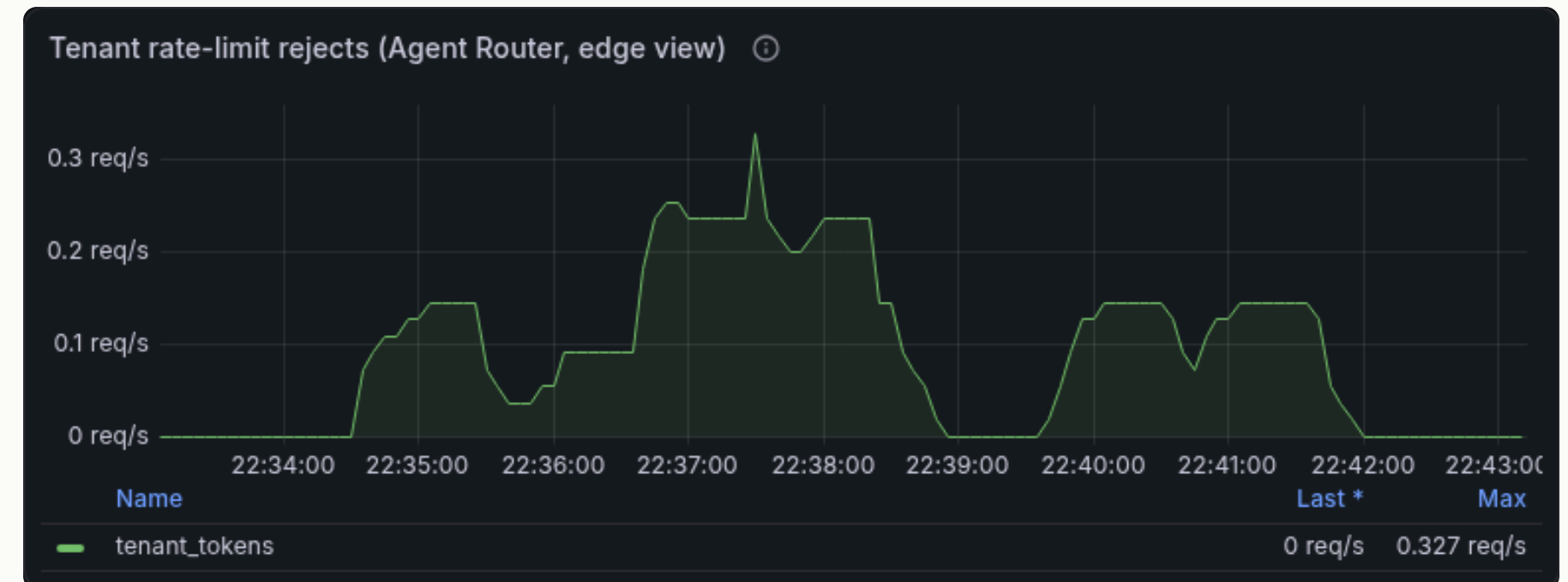
## WHAT WE MEASURED

Tenant test: one noisy tenant over its token budget, at 100% load.

**55 calls of the noisy tenant got 429 tenant\_tokens. No other tenant got a 429.**

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Dashboard 3 · Gateway + admission: the 429 tenant\_tokens rejects each second in the tenant test



## Redis and the queue

Redis holds the tenant counts. The queue is in the flow control of llm-d. Flow control is the admit part of llm-d.

# Place: prefix match first, then load

```
control/router/policy.yaml weights
control/router/render.py:50 render_epp_config()
```

## WHAT WE MEASURED

Stale-metrics test: one of two pods sends a frozen copy of its metrics from an empty moment, at 100% load. We measured its share of the work.

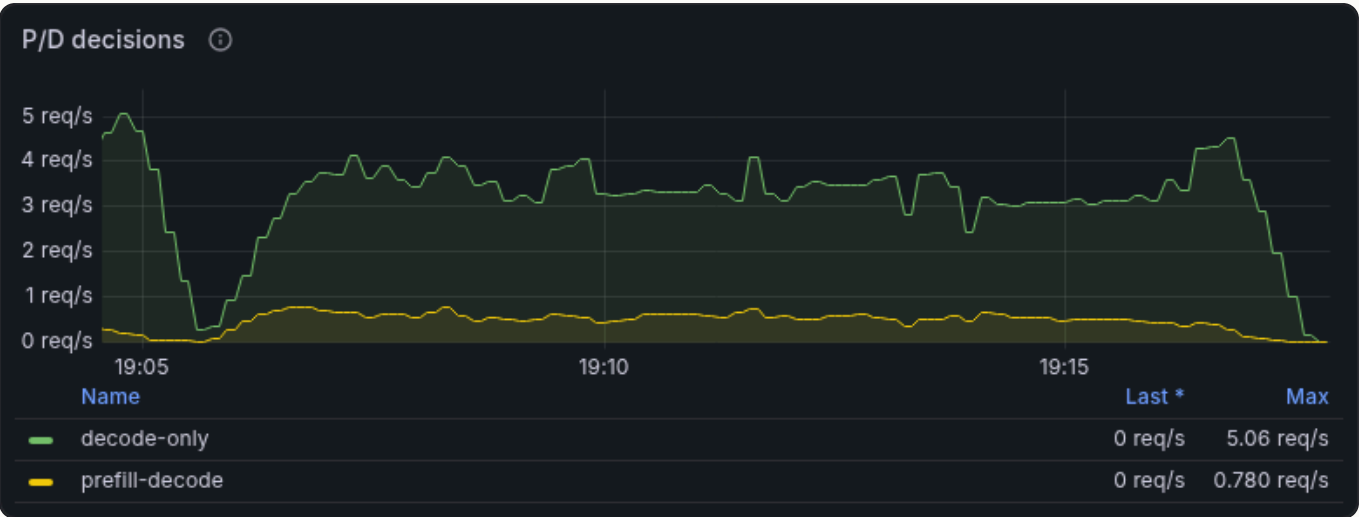
## 80% of the work

went to that pod. With real metrics, it got 51%.

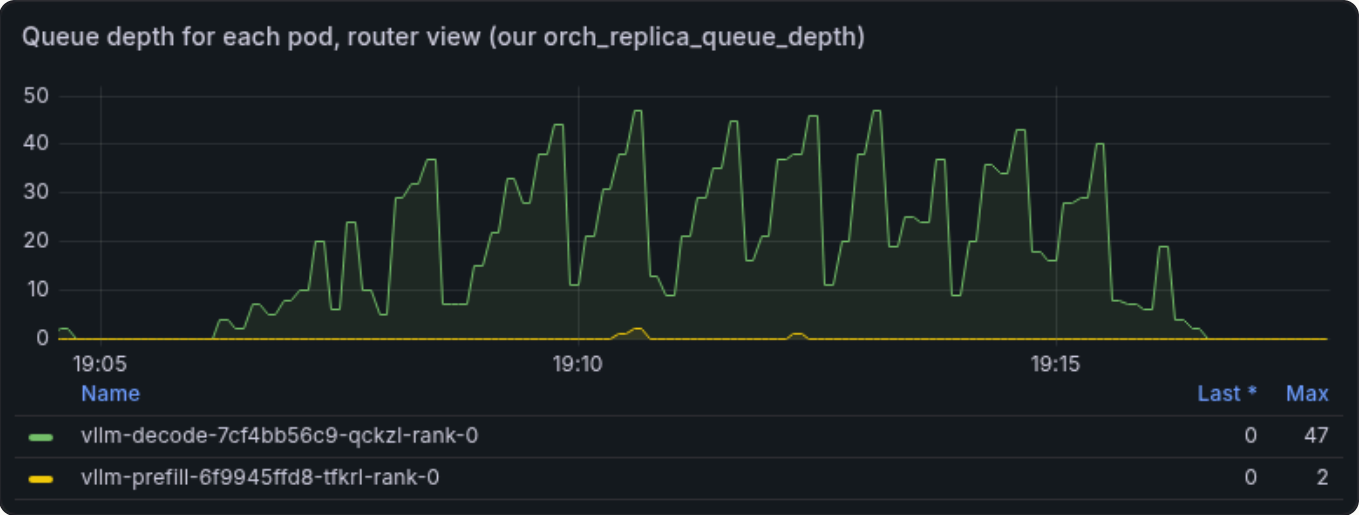
**What the data changed**

We cleared the prefix cache of one pod in a run. The llm-d scheduler did not act on the clear, and all 15 warm sessions went to that pod again and missed the cache once.

Dashboard 4 · Router: the decision for each call at 150% load: decode only, or split



Dashboard 5 · Queue depth by pod: decode up to 47, prefill up to 2



guard

stay or leave

admit

place

hop

warm

scale

# The hop: the KV moves through the LMCache server

```
control/router/policy.yaml
pd_non_cached_tokens: 2048
control/barrier/proxy.py:96 Barrier
tools/hop_records.py one record each hop
```

## WHAT WE MEASURED

Hop test: split requests of about 9,000 tokens. The time to the first token, by the path of the KV.

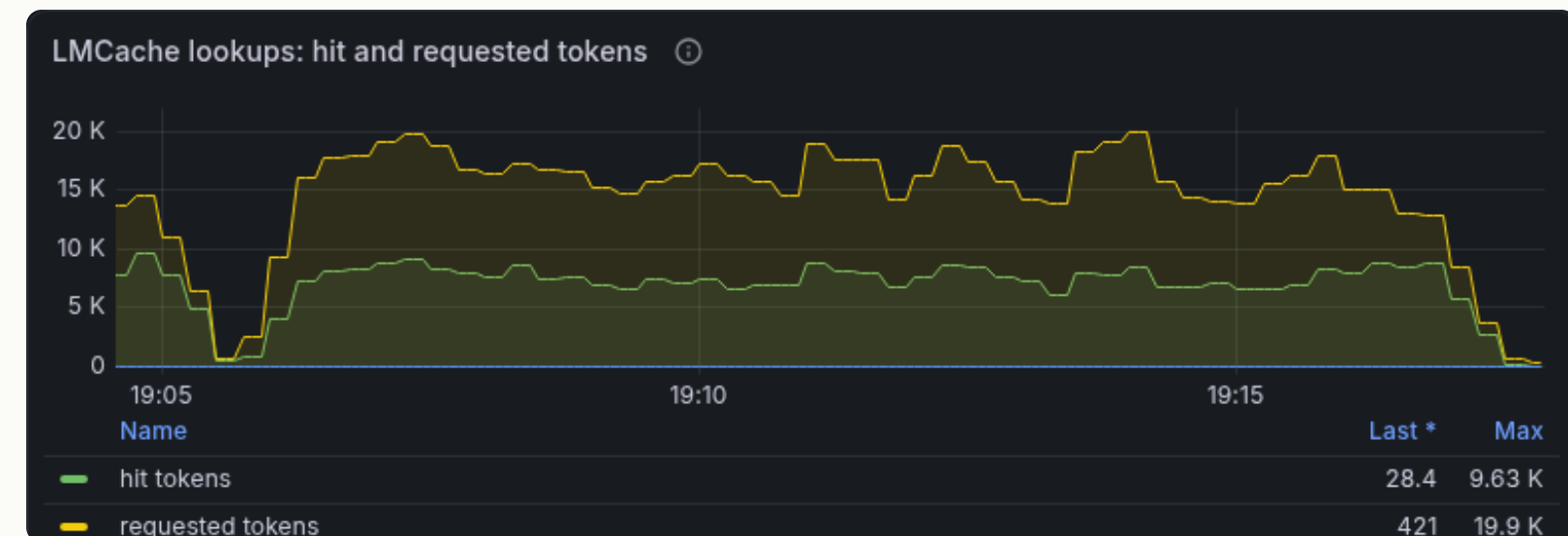
# 0.52 to 0.78 s

through the LMCache server, against 4.1 to 4.3 s through NIXL over TCP

### What the data changed

With no store barrier, the decode pod looked for the KV before the store ended: 0 hits in 10 tries. The barrier now holds the prefill answer until the store ends.

Dashboard 7 · Hop store: the tokens that the pods ask the LMCache server for, and the hit tokens



A hop record, from the Envoy log:

```
{"src": "10.42.1.21", "dst": "10.42.1.20",
"tokens": 6624, "prefix_tokens": 6400,
"backend": "lmcache"}
```

6,400 tokens are 25 full chunks of 256. The decode pod computes the KV of only the last 224 tokens.

Why not Mooncake: no Gemma 4 test, and no RDMA on our Lambda nodes.



guard

stay or leave

admit

place

hop

warm

scale

# A pod with its weights on the GPU is not warm yet

```
control/warm/logic.py:32 is_warm()  
control/warm/logic.py:36 next_ramp()
```

## WHAT WE MEASURED

Restart test at 70% load: we deleted one decode pod. The TTFT p95 of the calls on the new pod, in its first minute.

## 7.3 s against 10.9 s

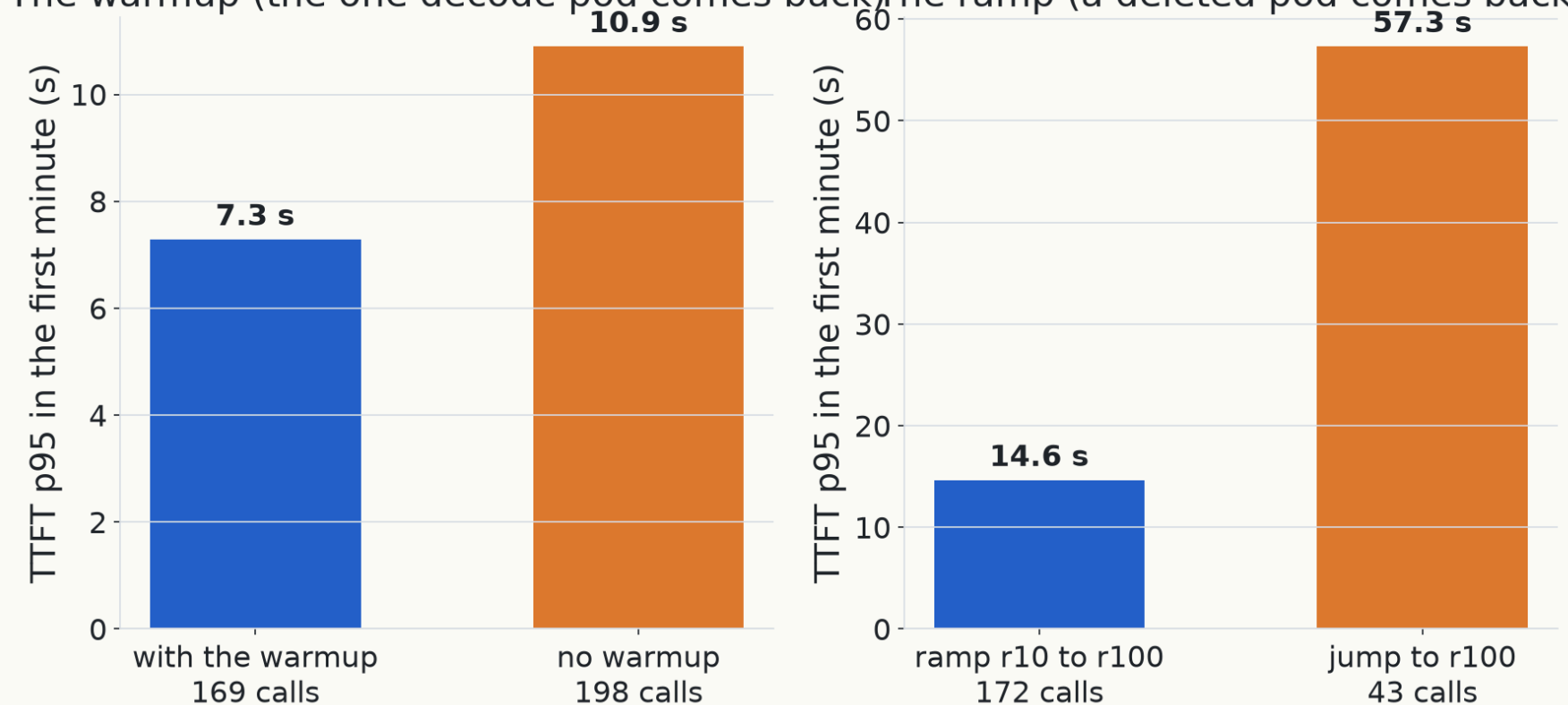
with the warmup routine, and with none. The warmup made the outage 15 s longer.

### What the data changed

The ramp is a score, not a cap. In the first 10 s, the new pod got 75% of the calls.

The TTFT p95 of the calls on the new pod, in its first minute (H100)

The warmup (the one decode pod comes back) The ramp (a deleted pod comes back)





# Scale: the planner names the pool

```
monitoring/rules.yaml:18 prefill: uncached
tokens
monitoring/rules.yaml:22 decode: running
KEDA one ScaledObject each pool
```

## WHAT WE MEASURED

Scale test on 8 × A100, one pool at a time: the time from the planner request to a new pod.

## 15 s to the new pod

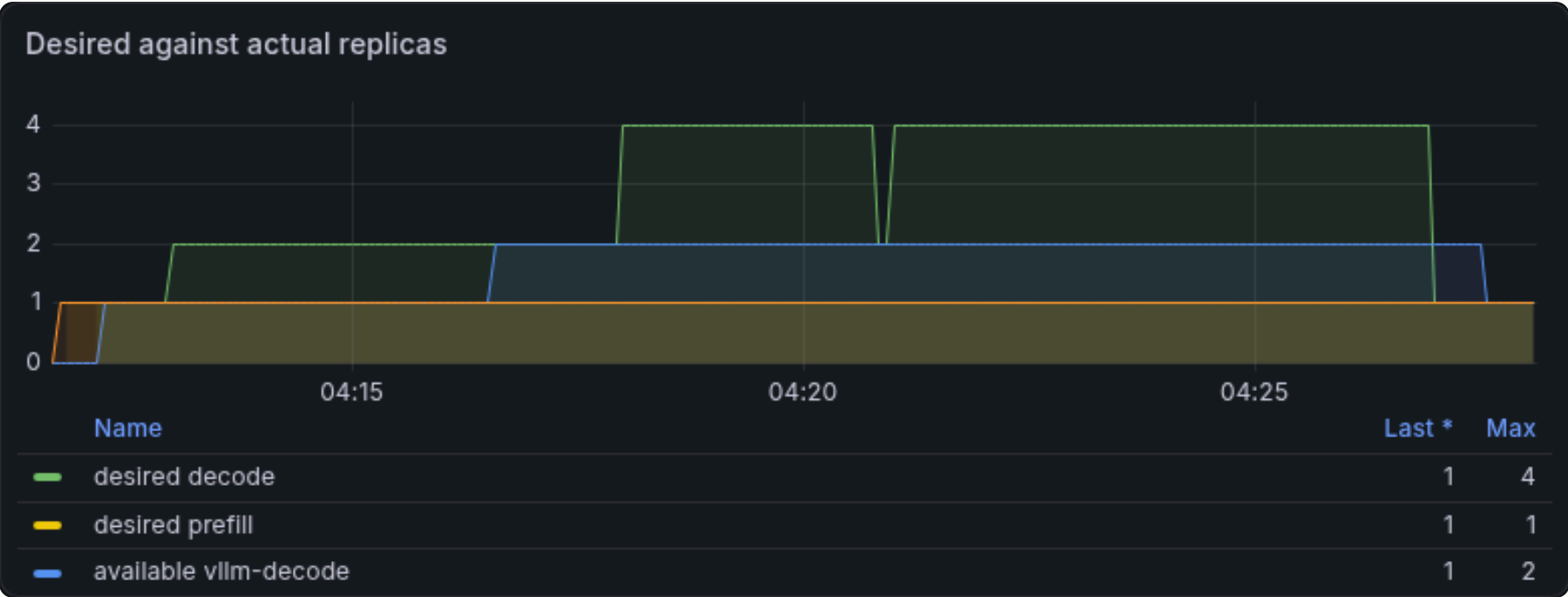
in both pools. The new pod was warm about 4 minutes later.

▶ Clip 2, 34 s, at 8 times speed: the replica panel

**What the data changed**

The prefill capacity value must match the GPU. With the H100 value, the planner asked for no second prefill pod on the A100.

Dashboard 8 · Pods / replicas / KEDA, in the scale test. Green: the decode pods that the planner asks for. Blue: the ready decode pods, at most 2 in this test.





# The decode pod was the limit, not prefill compute

| What we expected first | Result                                         |
|------------------------|------------------------------------------------|
| RAG: prefill compute   | Partly. Only for long prompts.                 |
| Agents: KV blocks      | Partly. The llm-d flow control held KV at 90%. |
| Not the weights        | Right. 30.4 of 71.7 GiB.                       |
| Not the interconnect   | Right for LMCache. Wrong for NIXL over TCP.    |
| Not the scheduler      | Wrong. The decode queue held 50 requests.      |

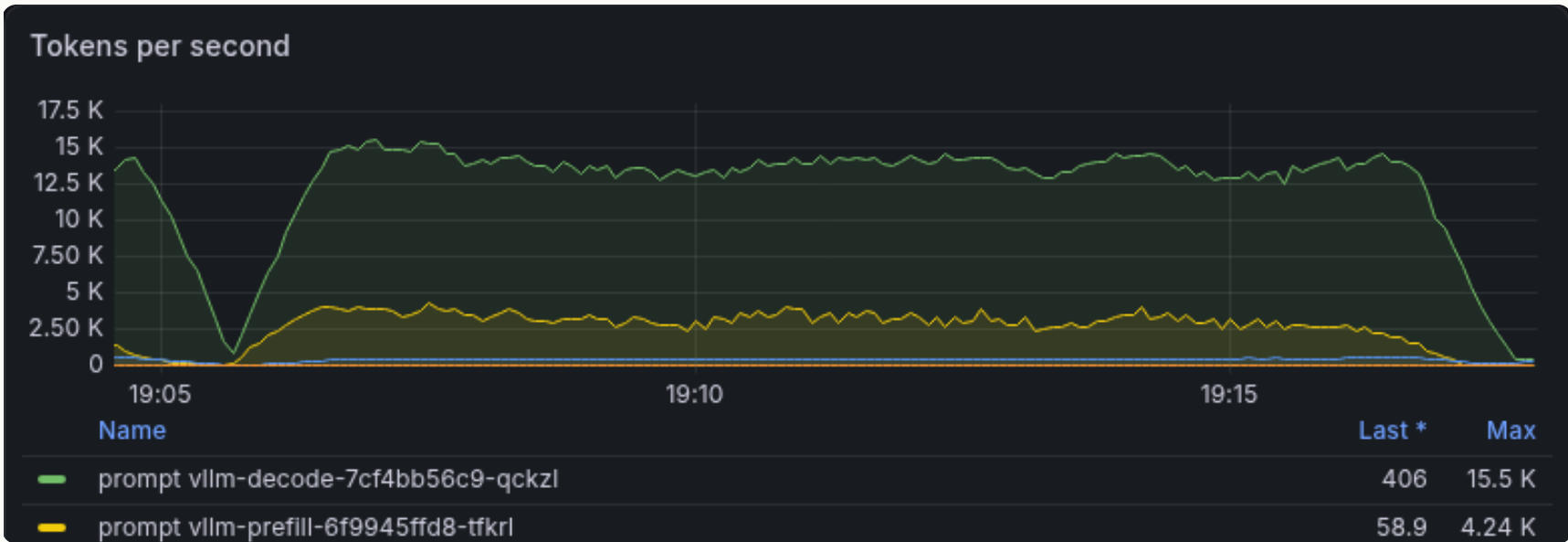
## WHAT WE MEASURED

At 100% load: the prompt tokens that each pod processed each second.

**16,200 against 4,550**

on the decode pod and on the prefill pod

Dashboard 6 · vLLM: the prompt tokens each second for each pod, at 150% load



# For our traffic, two colocated replicas beat a P/D split

## WHAT WE MEASURED

The same recorded app traffic at three loads, on two layouts: one prefill pod and one decode pod (P/D), or two colocated replicas. The TTFT p50 of the interactive calls.

## 0.84 s against 4.59 s

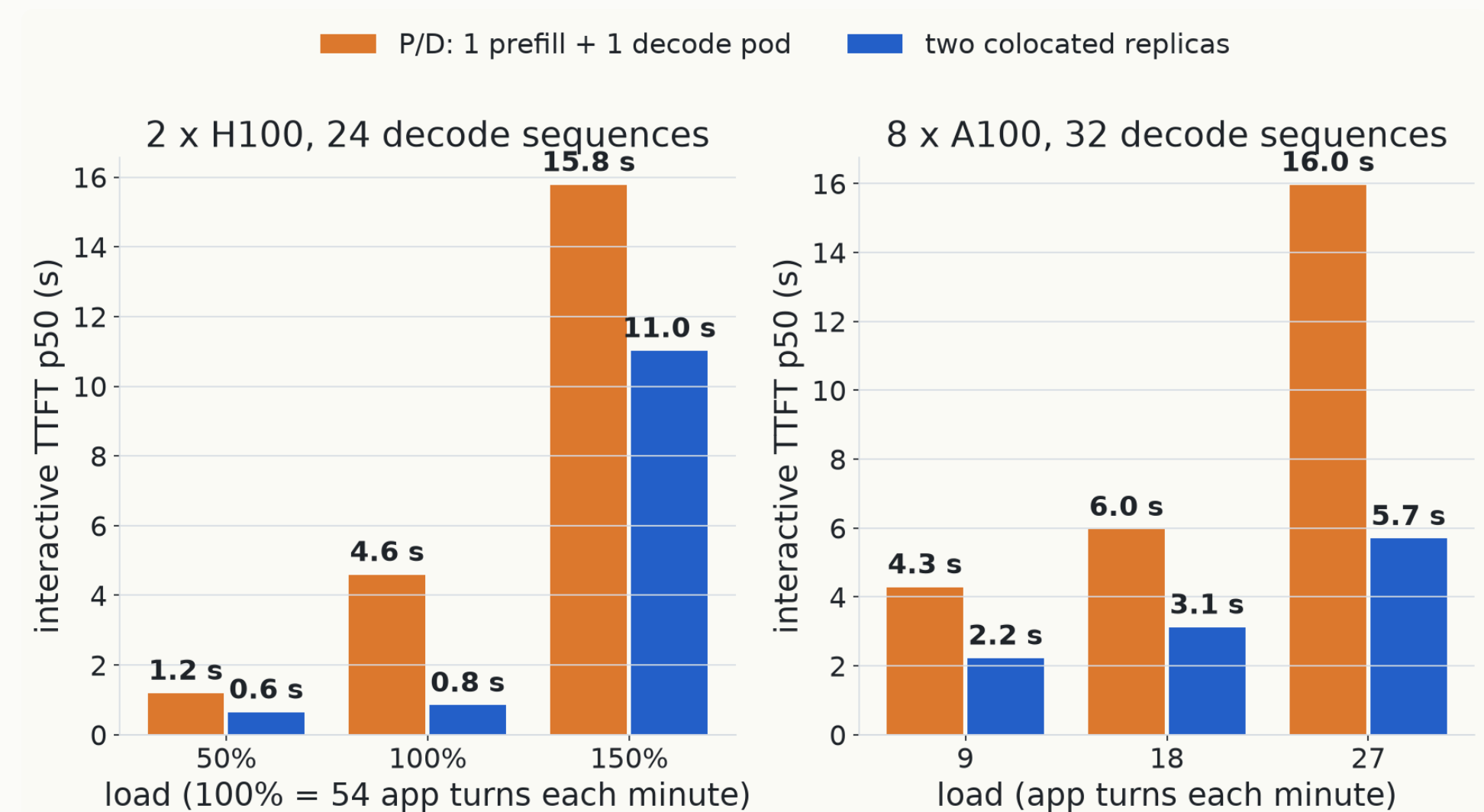
at 100% load on the H100: two colocated replicas against P/D. The A100 tests agree.

Why: only 10% to 13% of the calls split, so the decode pod runs the prefill of most prompts.

### What the data changed

We keep the split for long uncached prompts. A P/D layout needs two or more pods in each pool, or one restart stops all split calls.

The TTFT p50 of the interactive calls, by layout and load



# TTFT and ITL: P/D missed both SLOs at each load

## WHAT WE MEASURED

The same recorded app traffic on the two layouts: P/D, and two colocated replicas. TTFT: what the app saw, for the streaming calls (the answers). ITL: the p95 of each minute on the decode pod, the median over the run.

| Load | TTFT p95, P/D | TTFT p95, colocated | ITL p95, P/D | ITL p95, colocated |
|------|---------------|---------------------|--------------|--------------------|
| 50%  | 2.04 s        | 1.05 s              | 84 ms        | 36 ms              |
| 100% | 15.95 s       | 1.57 s              | 192 ms       | 50 ms              |
| 150% | 21.45 s       | 21.10 s             | 206 ms       | 129 ms             |
| SLO  | 1.5 s         | 1.5 s               | 50 ms        | 50 ms              |

One call at a time, in the gate test: TTFT 0.90 s, and ITL p95 20.5 ms.

TPOT at the client, the p50 of the calls: P/D 32 to 51 ms, colocated 25 to 41 ms. The engine has the same means.

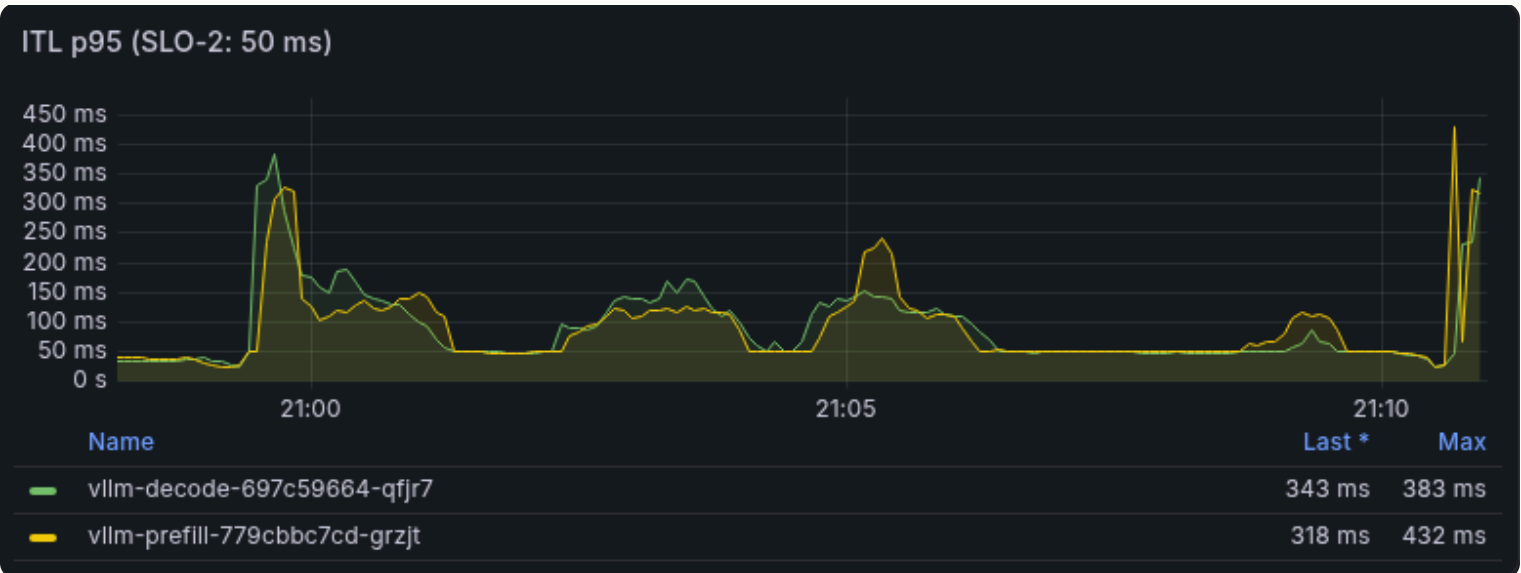
### Why the ITL of P/D is high

Most calls do not split, so the decode pod also computes their prompts. These prefill chunks share each step with the decode streams.

Dashboard 6 · vLLM: ITL p95, P/D at 100% load. About 200 ms most of the time.



Dashboard 6 · vLLM: ITL p95, two colocated replicas at 100% load. Near 50 ms most of the time, with bursts.



# TTFT at three points: under load, calls wait before the engine

| Layout and load | Client  | Gateway | Engine  | llm-d queue |
|-----------------|---------|---------|---------|-------------|
| P/D, 50%        | 2.04 s  | 1.87 s  | 0.80 s  | 0.00 s      |
| Colocated, 50%  | 1.05 s  | 0.94 s  | 0.72 s  | 0.00 s      |
| P/D, 100%       | 15.95 s | 15.67 s | 4.38 s  | 2.25 s      |
| Colocated, 100% | 1.57 s  | 1.35 s  | 0.90 s  | 0.00 s      |
| P/D, 150%       | 21.45 s | 21.34 s | 16.90 s | 9.53 s      |
| Colocated, 150% | 21.10 s | 20.79 s | 9.60 s  | 8.28 s      |

All values are p95. Client and gateway: the same streaming calls. Engine and queue: the p95 of each minute, the median over the run.

## WHAT WE MEASURED

Client: the load generator, to the first answer token.  
Gateway: edge, to the first byte. Engine: vLLM on the decode pod. Queue: the interactive band in llm-d.

### What the gaps mean

Client to gateway: 0.1 to 0.3 s, the relay. Gateway to engine: the guard, the llm-d queue, and the hop. With P/D at 100% load, this gap was 11 s.

### On the dashboards

Dashboard 6: the engine TTFT for each pod. Dashboard 3: the edge TTFT, but of all calls, so a call that does not stream counts its full answer. The client TTFT is in the load test logs.

# The handout questions: our answers and the evidence (1 of 2)

| Question                                                        | Our answer                                                                                                                              | File or scrape                                                                           |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| What is the app? Which tokens are common, and which are unique? | RAG and agent steps over the bookmarks. Shared: system prompt, tools, session history. Unique: question, chunks, pages.                 | metrics/cap-b1/envoy-access.log: an agent step finds 64% of its prompt in the cache.     |
| What dies at the guard, admit, place, and queue?                | Guard: unsafe prompts (400). Admit: a tenant over its budget (429). Queue: a wait past the time limit (503). Place: no ready pod (503). | docs/results.md: 22 of 22 attacks, 55 tenant 429 replies, 273 timeouts at 150% load.     |
| Where do I prevent work that will time out?                     | In the llm-d queue: 10 s for interactive and 120 s for batch. Edge sets each call to half of its time left.                             | control/router/policy.yaml:46 to :48, and the timeout_queue panel of dashboard 3         |
| Where do I protect KV?                                          | The llm-d flow control holds calls while the pods are full, so the KV stays near 90% or below. vLLM preemption is the last line.        | control/router/policy.yaml:55, and 0 preemptions in the raw scrape of a live engine      |
| Where do I prioritize interactive traffic?                      | Interactive goes before batch in the llm-d queue. Batch waits from 70% fullness, interactive only at 100%.                              | control/router/policy.yaml:46 to :54. At 100% load: 89 batch and 11 interactive sheds.   |
| Where do I limit one tenant, so that it cannot take the GPU?    | The Envoy AI Gateway counts the tokens and requests of each tenant in Redis. Over the budget: a 429 that never leaves.                  | control/router/policy.yaml:61. 55 429 replies for the noisy tenant, none for the others. |
| Where do I hop, and what is not copied?                         | llm-d splits a call at 2,048 or more uncached tokens. The decode pod computes again only the tokens after the last full chunk of 256.   | hops.jsonl of each P/D run: 2,304 of 2,371 tokens came from the cache.                   |



# The handout questions: our answers and the evidence (2 of 2)

| Question                                                                  | Our answer                                                                                                                  | File or scrape                                                                                 |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Where do I evict, and what becomes a ghost if I skip it?                  | vLLM and LMCache evict their oldest blocks. If llm-d misses a clear, the next call of each warm session misses: a ghost.    | metrics/e7b-*/kv-events.json and tools/ghost_probe.py: 15 of 15 sessions missed once.          |
| Where does the engine scheduler sit, against our admit, place, and queue? | Inside each vLLM pod, after our admit, place, and queue. We set its flags, and we do not change its code.                   | cluster/manifests/base/engine/vllm-decode.yaml:34 to :38                                       |
| What limited concurrency on this GPU for this app?                        | The decode pod. Most calls do not split, so it does their prefill and their decode, at 24 sequences.                        | Dashboard 6: 16,200 prompt tokens each second on the decode pod, 4,550 on the prefill pod.     |
| Four production alerts?                                                   | Interactive TTFT budget burn, a shed rate above 5%, KV pressure (above 92%, or preemptions), and hop failures.              | cluster/manifests/base/monitoring/rules.yaml:25 to :40                                         |
| If I scale, which pool: prefill tokens or decode slots?                   | Decode first, on running sequences against 60% of 24 slots. Prefill on uncached prefill tokens against 70% of its capacity. | rules.yaml:18 and :22. KEDA made a decode pod 15 s after the request.                          |
| What changes at 10 times the traffic?                                     | More decode capacity first: more pods, 32 sequences, and FP8 KV. More CPU RAM for LMCache.                                  | 32 sequences: TTFT p50 from 4.59 s to 2.10 s. FP8 KV: twice the tokens.                        |
| Which three knobs are the wrong next move?                                | More prefill pods, a longer queue, and a lower split threshold.                                                             | The prefill pod was not the limit. A longer wait still misses the goal. Each split costs more. |



# What the data changed in our design

## Colocated replicas

for our traffic. Split only long uncached prompts.

## A store barrier

for the hop: completion is not visibility.

## A cap for the ramp

not only a score.

## GPU values

from the GPU: the planner capacity and the warm baseline.

## Apply a cache clear

in llm-d, or each warm session misses the cache once.

10 times the traffic: more decode capacity first, 32 sequences and FP8 KV, and more RAM for LMCache. The wrong knobs: more prefill pods, longer queues, and a lower split threshold.

GPU cost: 237 USD of the 400 USD credit. The repo has the report, the notebooks, and each run.

## Questions?

[github.com/chetananand/learning-companion-serving](https://github.com/chetananand/learning-companion-serving)



# How llm-d places a call: the policy and the scorers

```
control/router/policy.yaml:15 policy: prefix_then_load
control/router/policy.yaml:36 weights: decode, prefill
control/router/render.py:50 render_epp_config()
```

| Scorer       | Decode | Prefill | What it scores                                    |
|--------------|--------|---------|---------------------------------------------------|
| prefix match | 3      | 3       | the part of the prompt in the KV cache of the pod |
| session      | 2      | -       | the pod that served the session before            |
| queue depth  | 2      | 2       | the requests that wait on the pod                 |
| KV use       | 2      | -       | the KV cache use of the pod                       |
| token load   | -      | 2       | the load of the pod in tokens                     |
| ramp         | 2      | 2       | a new pod gets 10%, 25%, 50%, then 100%           |

Filters first: only warm pods, and only pods with the role of the profile. Then the picker takes the pod with the highest total score.

### The policy

All runs used prefix\_then\_load. The other choices are least\_loaded and random. The llm-d scheduler has no p2c picker. With two pods in a pool, p2c is the same as least loaded.

### Queue depth: a scorer and an admit input

The queue scorer is in both profiles. The flow control also reads the queue depth: at 5 queued requests, a pod is full.

### WHAT WE MEASURED

Stale-metrics test: one pod sends a frozen copy of its metrics from an empty moment.

80% of the work

went to that pod. With real metrics, it got 51%.

# The queue questions: our answers and the proof

| Question                                                              | Our answer                                                                                                                                             | What we measured                                                                               |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Who waits in our queue, and who in the vLLM queue?                    | Ours is in the llm-d flow control: a call waits there before the pick, while the pods are full. In vLLM, a call waits inside the pod for a batch slot. | At 150% load: up to 48 interactive and 24 batch calls in ours, and up to 50 in the decode pod. |
| Waiting, running, preempted?                                          | vLLM has no swap: it preempts by computing the KV again. The decode pod runs at most 24 calls.                                                         | Decode: up to 24 running and 50 waiting. 0 preemptions in all runs.                            |
| Queue depth of each pod, by mix?                                      | We read the queue of each pod from llm-d and from vLLM, for each traffic mix (dashboard 5).                                                            | 6 or fewer for most mixes. Up to 36 on the decode pod with the mixed app traffic.              |
| A 32K retrieve and a short agent step are both ready. Who goes first? | The short step. The split sends the long prompt to the prefill pod. Without the split, vLLM computes it in chunks.                                     | The short step was first in all rounds: 0.25 s with the split, 1.0 to 1.26 s without.          |
| PagedAttention or the prefix cache: which saved memory?               | Neither changed the KV use at our load. The prefix cache saved compute. FP8 KV saved memory.                                                           | KV use 13% with and without the prefix cache, 5.5% with FP8. 44% of prompt tokens were hits.   |
| Engine flags for chunked prefill and batching?                        | Decode: 24 sequences, chunks of 2,560 tokens, so a short prefill fits one chunk. Prefill: 8 sequences, chunks of 16,384.                               | With 32 decode sequences, the TTFT p50 fell from 4.59 s to 2.10 s.                             |
| KV full after admit: the door or preemption?                          | The door. The llm-d flow control holds calls while the pods are full, so vLLM did not preempt.                                                         | Soak test: the KV use stayed at 90% or less, with 0 preemptions.                               |
| The client is gone: who frees the KV?                                 | vLLM. Envoy closes the stream, and vLLM stops the call and frees its KV blocks.                                                                        | Envoy closed all 46 aborted streams at the client time.                                        |
| A pod returns: 100% at once, or a ramp?                               | A ramp: 10%, 25%, 50%, then 100% of the score, while the TTFT p99 of the pod holds.                                                                    | First-minute TTFT p95: 14.6 s with the ramp, 57.3 s with a jump.                               |

# The hop record, and a cold pod against a warm pod

Same pod: llm-d picks only the decode pod. The KV is already there, so there is no hop and no record.

Two pods: one hop record for each split call, from the Envoy log:

```
{"src": "10.42.1.21", "dst": "10.42.1.20",  
"tokens": 2371, "prefix_tokens": 2304,  
"backend": "lmcache", "first_byte_ms": 496}
```

The field src is the prefill pod, and dst is the decode pod. The field prefix\_tokens is the part of the prompt that the decode pod did not compute. The field first\_byte\_ms is the TTFT.

**The warmup routine**  
Our system prompts, the shapes of our app, one split call through the prefill pod, and a 4,000-token probe. The pod is warm only if the probe TTFT is at most 1.5 times the warm baseline.

## WHAT WE MEASURED

Restart test: we deleted the one decode pod at 70% load. The TTFT of the calls on the new pod, in its first minute.

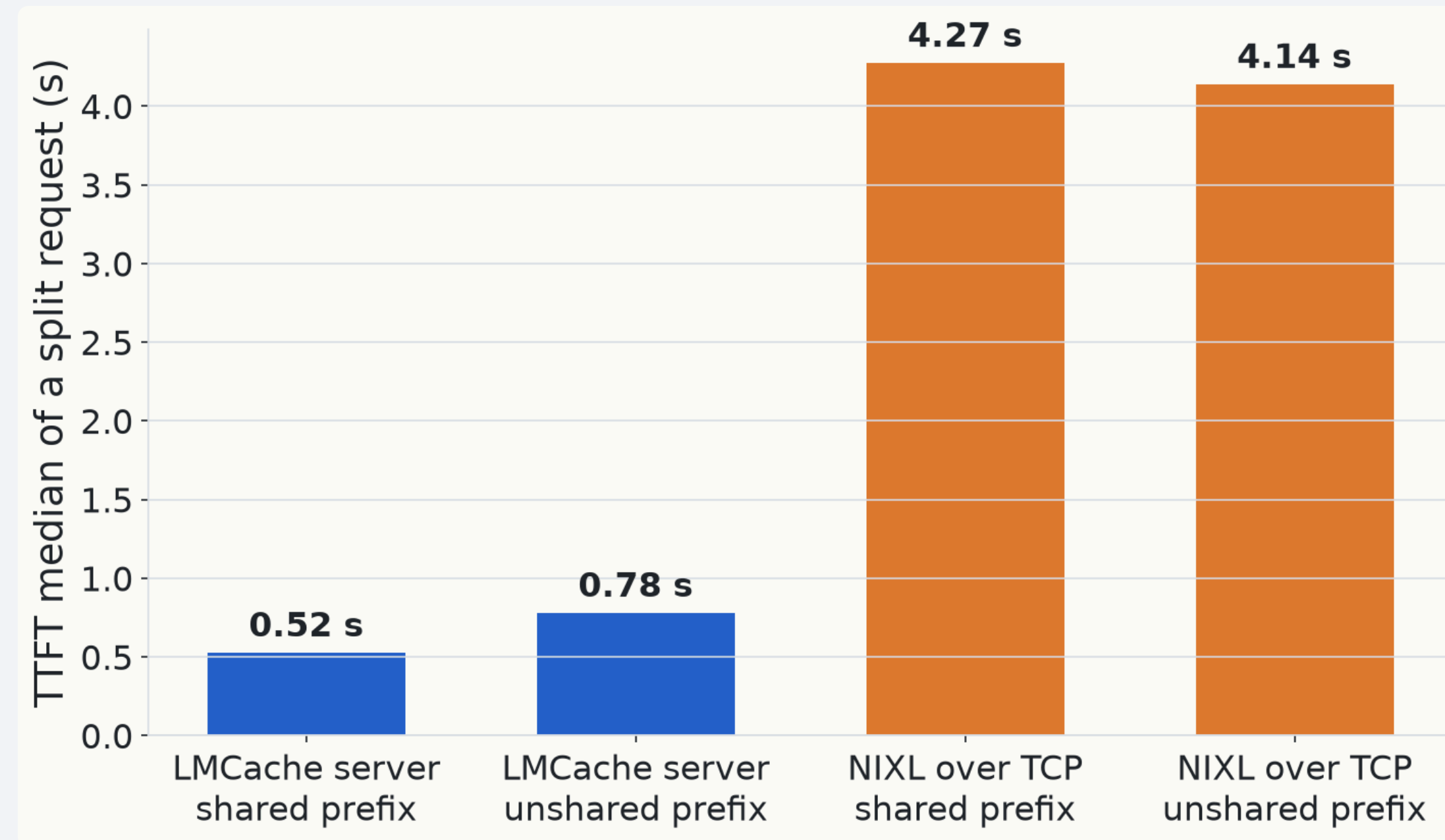
|          | No warmup | Warmup |
|----------|-----------|--------|
| TTFT p95 | 10.9 s    | 7.3 s  |
| TTFT p50 | 1.29 s    | 1.03 s |
| Outage   | 275 s     | 290 s  |

Minutes 2 to 4: a TTFT p95 near 12.7 s in both arms. The warmup helps only the first minute.

## Appendix

# The hop: the LMCache server against NIXL

The time to the first token of a split request of about 9,000 tokens



# The hop at production scale: what we keep, what we change

**Keep**

The llm-d scheduler decides, and the sidecar runs the two steps. The one-token prefill request: the smallest vLLM request that runs a full prefill.

**WHAT WE MEASURED**

The store barrier on the A100 node, under load. It waits for every store that LMCache had at that time, with a cap of 0.5 s.

| When                                    | Holds | Hit the 0.5 s cap     |
|-----------------------------------------|-------|-----------------------|
| P/D load tests                          | 237   | 170 (72%)             |
| End of session, prefill pod 1           | 171   | 163 (95%)             |
| End of session, prefill pod 2           | 68    | 67 (99%)              |
| Start of session, one request at a time | 3     | 0 (0.19 s on average) |

**A production hop needs**

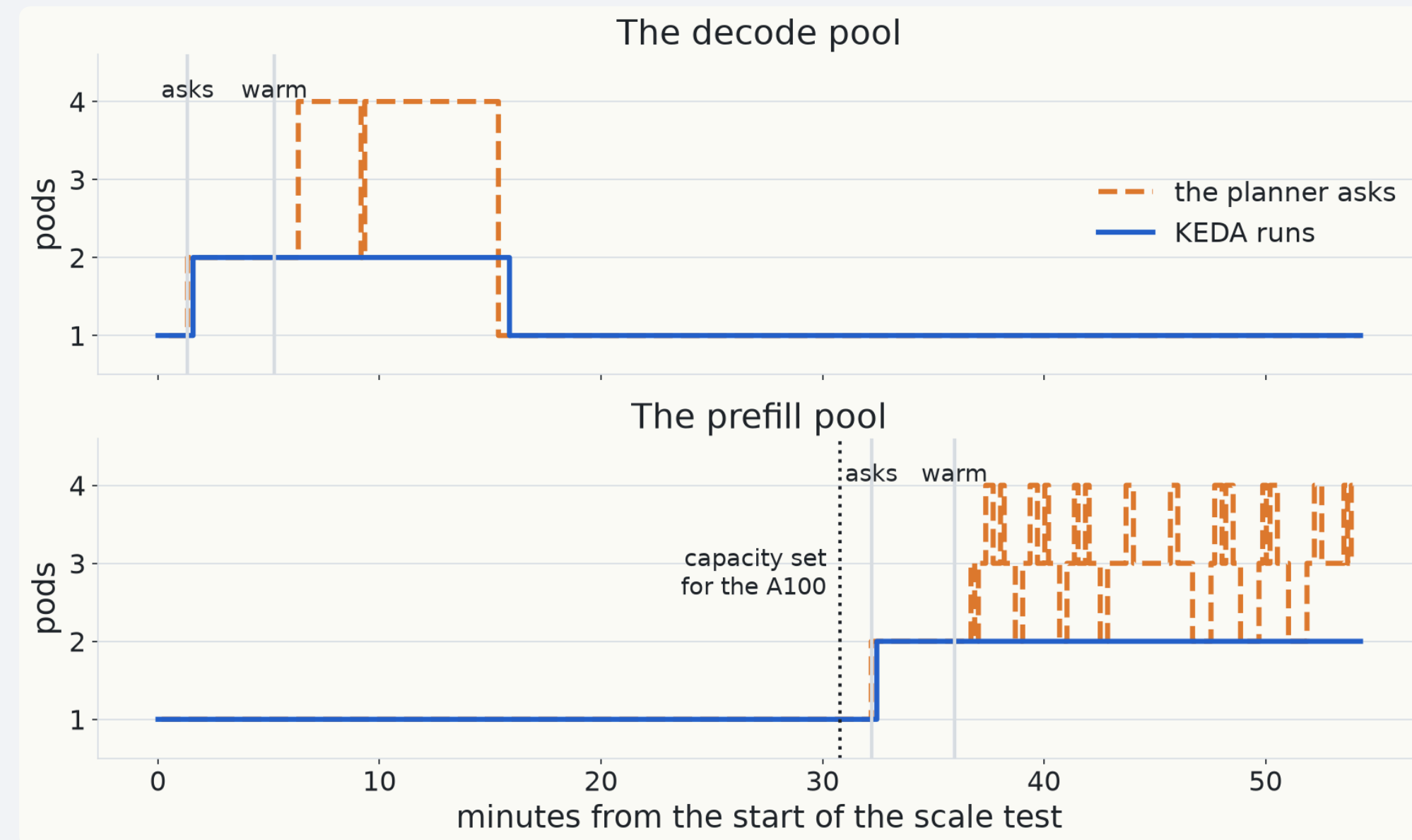
1. A store signal for each request, in place of the global store counters.
2. RDMA between nodes, with NIXL or a distributed KV store. Our LMCache path works only inside one node, and NIXL over TCP took about 4 s.
3. Two or more pods in each pool, and a fallback: if the prefill request fails, the decode pod does the prefill itself.
4. mTLS between pods. Our sidecar ran with plain HTTP.
5. P/D only where long uncached prompts are common. For our traffic, two colocated replicas were faster.



## Appendix

# The scale test: the planner against KEDA, in each pool

The pods that the planner asks for, and the pods that KEDA runs. One pool at a time.



# Traps that the handout names, and what our design does

| The trap                                                   | What our design does, and the proof                                                                                                |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| A benchmark at batch 8 is the production SLO.              | Our SLO comes from the app: interactive TTFT p95 at most 1.5 s up to 8K tokens, on a warm pod, with recorded app traffic.          |
| The cache is full, so add a replica of the same size.      | First make the KV smaller (FP8 KV: twice the tokens) and keep the prefixes in LMCache. Then scale the pool that the planner names. |
| NCCL or NIXL in this repo moves the KV.                    | The LMCache connector in vLLM moves the KV. Our code records the hop and holds the prefill answer until the store ends.            |
| A replica is ready when the weights are on the GPU.        | A pod is warm only after the warmup and a probe. The warmup cut the first-minute TTFT p95 from 10.9 s to 7.3 s.                    |
| The overflow is another API, with no model and no limiter. | qwen3.8-27b on the Superlinked API, only for an interactive 503 or 529. A Redis limiter caps the requests, tokens, and cost.       |
| I wrote my own vLLM scheduler in the gateway.              | No. vLLM schedules inside each pod, and we only set its flags. The gateway decides what enters, the wait order, and the pod.       |
| A 429 that left the cluster.                               | The leave gate keeps each 429. At 150% load, it let only 124 interactive 503 calls go.                                             |
| The gateway fixed OOM.                                     | No. vLLM manages the GPU memory. The gateway keeps the load below preemption, and a prompt that is too long gets a 413.            |
| RAG is a third phase.                                      | No. The search runs outside the LLM, with SIE and Qdrant. Its chunks are prompt tokens: the engine sees only prefill and decode.   |
| Wall seconds across models, with no token counts.          | We compare models for each token: KV bytes and the time between tokens. Each load test replays the same traffic in each arm.       |
| The TTFT of a cold pod as the SLO.                         | Our SLO runs use warm pods. The first minute of a new pod is a separate measure: 10.9 s cold, 7.3 s warm.                          |



# A raw /metrics scrape of a live engine

```
# the vllm-decode pod
vllm:num_requests_running{engine="0",model_name="companion"} 1.0
vllm:kv_cache_usage_perc{engine="0",model_name="companion"} 0.01538461538461533
vllm:num_preemptions_total{engine="0",model_name="companion"} 0.0
vllm:prompt_tokens_total{engine="0",model_name="companion"} 5.426828e+06
# edge
orch_shed_total{code="400",reason="prompt_injection",stage="guard"} 7.0
orch_shed_total{code="503",reason="timeout_queue",stage="flow_control"} 138.0
orch_overflow_refused_total{why="no_provider"} 89.0
# the store barrier
barrier_hold_seconds_sum 85.915873
barrier_hold_seconds_count 171
```

Saved from the live A100 node before we stopped it. The report has more scrapes.

# Faults that we found and fixed

| Fault           | What happened                                         | Fix                 |
|-----------------|-------------------------------------------------------|---------------------|
| hop race        | The decode lookup ran before LMCache ended the store. | the store barrier   |
| Envoy buffer    | Envoy gave 413 for prompts of about 12K tokens.       | a 4 MiB buffer      |
| gateway timeout | The AI Gateway stopped each request at 60 s.          | a timeout of 300 s  |
| lost pod label  | The warm-controller series lost the pod label.        | honorLabels: true   |
| prefill restart | The one prefill engine restarted under overload.      | not fixed: two pods |
| cache reset     | The cache reset said 200 with success false.          | a forced reset      |
| hop records     | The hop records named NIXL for each run.              | the real connector  |

Appendix

# The demo questions: 8 of 12 on the H100, 6 of 12 on the A100

| Run              | Passed           | What failed                                           |
|------------------|------------------|-------------------------------------------------------|
| H100, 2026-09-30 | 8 of 12          | 4 questions in the fact check                         |
| A100, 2026-10-01 | 6 of 12          | 5 questions in the fact check, and 1 slow first token |
| After the fixes  | the OCR question | Claim checks still time out or find no evidence.      |

The failures are in the fact check, not in the serve path.

# The cost of each GPU block

| Date (PDT) | Shape                            | Hours | USD    |
|------------|----------------------------------|-------|--------|
| 2026-09-27 | 2 × H100 SXM (first engine test) | 0.37  | 3.07   |
| 2026-09-28 | 2 × A6000 (first platform test)  | 1.92  | 4.17   |
| 2026-09-29 | 1 × H100 SXM (node 2)            | 11.40 | 48.91  |
| 2026-09-29 | 2 × H100 SXM (node 1)            | 9.85  | 82.56  |
| 2026-09-30 | 4 × H100 SXM (no node 2 came)    | 0.75  | 12.20  |
| 2026-10-01 | 8 × A100 80 GB (one node)        | 3.88  | 86.51  |
| Total      | of the 400 USD credit            |       | 237.43 |